PogranichnyZashchitnik at ArA-DF 2026: WavLM Large with LoRA and One-Class Softmax for Arabic Speech Deepfake Detection

Minh-Hoang Le
University of Information Technology, Ho Chi Minh City, Vietnam
Vietnam National University, Ho Chi Minh City, Vietnam
24520542@gm.uit.edu.vn

Abstract

This paper describes the system submitted by team **PogranichnyZashchitnik** to the ArA-DF 2026 shared task on Arabic Audio Deepfake Detection. We build on **WavLM Large** as a self-supervised feature extractor, fine-tuned with **Low-Rank Adaptation (LoRA)** so that only a small fraction of parameters need updating. Representations from all 25 hidden layers are blended via a **learnable weighted sum**, pooled into utterance-level embeddings with **Attentive Statistics Pooling**, and classified by an **OC-Softmax** head that treats bonafide speech as the target class. Combined with an augmentation pipeline of codec compression, additive noise, and reverberation, this single system achieves **12.54%** EER on Track 1 (Dialect Generalization, 14th/15) and **5.66%** EER on Track 2 (Acoustic Robustness, 8th/14), outperforming the organizer baseline on both tracks. Post-evaluation analysis shows that utterances shorter than two seconds are disproportionately difficult, and that a non-trivial fraction of spoofed samples are scored as confidently bonafide under dialect mismatch.

1 Introduction

As text-to-speech and voice conversion tools become increasingly accessible, the risk of audio deepfakes - synthetic speech designed to impersonate real speakers - grows alongside them. Most anti-spoofing research has concentrated on English through the ASVspoof series (Yamagishi et al., 2021; Wang et al., 2024), leaving many languages with limited coverage.

The ArA-DF 2026 shared task turns the spotlight on **Arabic**, a language whose many regional dialects and variable recording conditions make deepfake detection especially challenging. Participants are asked to classify each audio clip as bonafide or spoofed under two settings: Track 1 tests whether a detector still works when the dialect or speaker was never seen during training, while Track 2 probes robustness to real-world degradations like background noise, lossy compression, and re-recording.

Our approach rests on three ideas. First, we draw on WavLM Large (Chen et al., 2022) for its rich, pretrained speech representations, adapting it efficiently through LoRA (Hu et al., 2022) so that fewer than 2M parameters are actually trained. LoRA has already shown promise for spoofing detection in English (Wang et al., 2023; Laakkonen et al., 2025); we apply it here to Arabic. Second, we use OC-Softmax (Zhang et al., 2021) - a one-class objective that models only what bonafide speech looks like, rather than trying to carve out a decision boundary between two classes - on the premise that spoofing attacks are open-ended and a closed-set classifier will always be one step behind. Third, we lean heavily on data augmentation (codec compression, environmental noise, reverberation) to bridge the gap between clean training data and messy real-world conditions.

We submit a single model to both tracks without any track-specific tuning. Our contribution is not methodological - each component has established precedents - but rather the combination and evaluation of these techniques for Arabic, along with an honest analysis of where the system succeeds and where it falls short. Two findings stand out from our post-evaluation analysis: (1) the system handles acoustic degradation far better than dialectal mismatch, and (2) very short clips ($<$2 s) remain a weak spot, with error rates roughly doubling.

2 Task and Data

Task. Each input is an Arabic audio clip; the system must decide whether it is *bonafide* (label 1) or *spoofed* (label 0). The primary metric is Equal Error Rate (EER).

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 1491–1495
October 28-29, 2026 ©2026 Association for Computational Linguistics

Split	Total	Bonafide	Spoof
Train	22,500	7,500	15,000
Track 1 Test	144,210	36,903	107,307
Track 2 Test	127,746	31,490	96,256

Table 1: Dataset statistics.

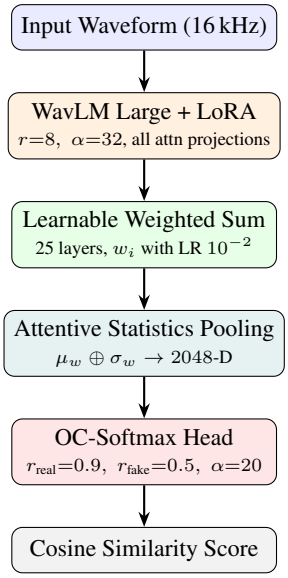


Figure 1: System architecture. The same model is used for both tracks.

Tracks. Track 1 (**Dialect Generalization**)[1] tests detection on dialects and speakers absent from the training data. Track 2 (**Acoustic Robustness**)[2] introduces realistic degradations - noise, codec artifacts, re-recording - at test time.

Data. Table 1 summarizes the ArA-DF 2026 dataset.[3] The training split covers three Arabic dialect groups (Egyptian, Gulf, and MSA). Both test sets skew heavily toward spoofed samples (roughly 3:1).

3 System Description

The system has three stages: a WavLM Large frontend adapted with LoRA, an Attentive Statistics Pooling layer, and an OC-Softmax classification head. Figure 1 gives an overview.

3.1 WavLM Large Frontend

WavLM Large[4] (Chen et al., 2022) stacks a CNN feature encoder and 24 Transformer layers (hidden size 1024). Rather than taking only the final layer, we let the model decide which layers matter by learning a **weighted sum** over all 25 hidden states:

$$\mathbf{x}_{\text{agg}} = \sum_{i=0}^{24} \text{softmax}(\mathbf{w})_i \cdot \mathbf{h}_i \tag{1}$$

The weight vector $\mathbf{w} \in \mathbb{R}^{25}$ starts at uniform values and is trained at 10^{-2}, 100× the learning rate of other parameters, so it can adapt quickly.

To keep fine-tuning lightweight, we insert **LoRA** modules (Hu et al., 2022) into every self-attention projection (`q_proj`, `k_proj`, `v_proj`, `out_proj`) with rank r=8, scaling α=32, and dropout 0.1. The rest of WavLM stays frozen, leaving roughly **1.7M trainable parameters** out of 316M total.

3.2 Attentive Statistics Pooling

Attentive Statistics Pooling (ASP) (Okabe et al., 2018) converts the variable-length frame sequence into a single utterance vector. A small attention network scores each frame's relevance; those scores weight both the mean and standard deviation across time, and the two are concatenated into a 2048-dimensional embedding (2×1024). In practice, this lets the model focus on the frames that carry the most spoofing evidence while downweighting silence or uninformative regions.

3.3 OC-Softmax Classification Head

A standard binary classifier learns boundaries for both classes, but spoofed speech is an open set - new generation methods appear constantly, and a model trained against today's attacks may miss tomorrow's. OC-Softmax (Zhang et al., 2021) sidesteps this by modeling only the bonafide class. It learns a center vector $\mathbf{c}$ and scores each input by cosine similarity:

$$s = \frac{\mathbf{e}}{||\mathbf{e}||} \cdot \frac{\mathbf{c}}{||\mathbf{c}||} \tag{2}$$

Training pulls bonafide embeddings toward $\mathbf{c}$ (target r_{real}=0.9) and pushes spoofed ones away (r_{fake}=0.5), scaled by α=20. At test time, a high score means the model thinks the clip sounds like genuine speech.

4 Experimental Setup

4.1 Data Augmentation

We found augmentation to be the single most impactful ingredient for Track 2, so we apply a fairly

[1] https://www.codabench.org/competitions/17138/
[2] https://www.codabench.org/competitions/17139/
[3] https://huggingface.co/datasets/ArabicSpeech/ArA-DF-2026
[4] https://huggingface.co/microsoft/wavlm-large

Augmentation	Prob.	Details
MP3 compression	15%	FFmpeg codec
AAC compression	15%	FFmpeg codec
OGG compression	20%	FFmpeg codec
SpecAverage	50%	2 bands, width=10
Additive noise	50%	MUSAN, SNR 10–25 dB
Reverberation	50%	RIRS_NOISES
SpecAugment	—	p_t=p_f=0.05, drop=0.1

Table 2: Online data augmentation pipeline applied to waveforms during training.

aggressive pipeline directly to the waveforms during training (Table 2). Codec compression through FFmpeg (MP3, AAC, OGG) is meant to mimic what happens when audio passes through social media platforms or messaging apps. Additive noise sampled from the MUSAN corpus (Snyder et al., 2015) and convolution with room impulse responses from RIRS_NOISES (Ko et al., 2017) simulate noisy, reverberant environments. On top of this, WavLM's built-in SpecAugment (Park et al., 2019) masks random time–frequency regions.

4.2 Training Details

Audio clips are truncated to at most 9.0 s during training to keep memory usage manageable. We optimize with AdamW (learning rate 10^{-4}, weight decay 10^{-5}) for all parameters except the layer aggregation weights, which use a learning rate of 10^{-2}. With a per-device batch size of 4 and 4 gradient accumulation steps (effective batch 16), we train for 5 epochs under a flat learning rate. The checkpoint at epoch 4, which gave the lowest EER on the validation split (0.63%), is kept for submission. Training uses mixed-precision (bfloat16) on a single laptop GPU.

4.3 Inference

Each test file is loaded up to 9.0 s and split into non-overlapping **4.0 s chunks**; anything shorter is zero-padded to 4.0 s. Chunks are fed through the model in sub-batches of 16, and the resulting embeddings are averaged per file before the OC-Softmax head produces a final score:

$$\mathbf{e}_{\text{file}} = \frac{1}{N} \sum_{n=1}^{N} \mathbf{e}_{\text{chunk},n} \tag{3}$$

No ensembling or track-specific post-processing is applied.

Track	EER (%)	Rank	Baseline
1 (Dialect)	12.54	14/15	14.54
2 (Acoustic)	5.66	8/14	27.11

Table 3: Official test results. Baseline: Wav2Vec2 XLS-R 300M + AASIST.

Duration	T1 EER (%)	T2 EER (%)
0–2 s	29.83	16.34
2–4 s	16.03	8.09
4–6 s	12.13	4.95
6–8 s	12.20	4.75
8–10 s	12.00	5.02
10–30 s	12.18	5.84
Overall	**12.54**	**5.66**

Table 4: EER by utterance duration. Clips shorter than 2 s are dramatically harder.

5 Results and Analysis

5.1 Official Results

Table 3 shows our leaderboard scores next to the organizer baseline (Wav2Vec2 XLS-R 300M + AASIST). We beat the baseline on both tracks, but the margin is far larger on Track 2, an EER of 5.66% versus the baseline's 27.11%. This gap strongly suggests that our augmentation pipeline does a good job of matching the kinds of degradation present in the Track 2 test set, while dialectal variation in Track 1 calls for something more than signal-level augmentation.

5.2 Effect of Utterance Duration

Breaking EER down by utterance length (Table 4) reveals a striking pattern: clips under 2 s are far harder to classify, with EER jumping to 29.8% on Track 1 and 16.3% on Track 2. This makes sense given our chunking strategy - a 1.5 s clip gets padded to 4 s, so more than half the input is silence, and the model has little genuine signal to work with. Once clips exceed 4 s, performance stabilizes and roughly matches the overall average, confirming that the chunk window is well-sized for typical-length recordings.

5.3 Score Distribution

Table 5 summarizes the OC-Softmax cosine scores on each test set. The two classes are reasonably well-separated on both tracks, but Track 1 has a troubling tail: **11.5% of all spoofed samples** (12,368 out of 107,307) receive scores of 0.98 or higher - essentially indistinguishable from bonafide

Track	Class	Mean	Std	Min	Max
T1	Bonafide	0.962	0.112	−0.110	0.988
	Spoof	0.417	0.325	−0.315	0.988
T2	Bonafide	0.950	0.126	0.004	0.988
	Spoof	0.189	0.249	−0.406	0.988

Table 5: OC-Softmax cosine score statistics on each test set. Higher scores indicate bonafide.

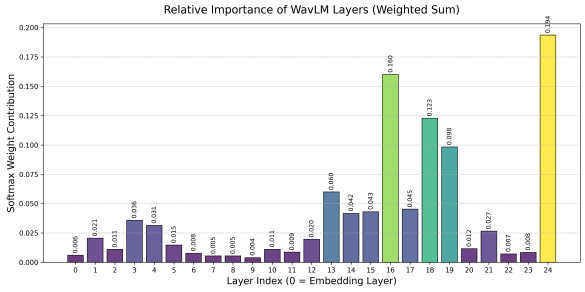


Figure 2: Learned layer weights (softmax-normalized). The model relies most on the upper Transformer layers.

speech. Among these, 3,471 hit the maximum model output of 0.988. These are presumably high-quality deepfakes from dialectal contexts the model never saw during training: if the model has only learned what Egyptian or Gulf bonafide speech "looks like," a convincing Levantine deepfake can slip right past. Track 2 shows much cleaner separation, with a wider gap between class means and far fewer false positives near the decision boundary.

5.4 Layer Weight Analysis

Figure 2 plots the learned layer weights after training. The model leans heavily on the **top Transformer layers**: layers 24, 16, and 18 together claim 48.7% of the total weight, while every layer below 10 contributes under 2%. This is consistent with speaker verification work (Desplanques et al., 2020), where upper layers capture the kind of abstract speaker and style information most useful for spotting synthetic artifacts.

Limitations

Several limitations are worth noting. We do not perform any dialect-aware training or adaptation, which is likely the main reason Track 1 lags behind. We also do not present a component-level ablation study: each design choice (WavLM, LoRA, ASP, OC-Softmax) is motivated by prior work, but we have not isolated their individual contributions on this dataset. Training runs for only 5 epochs, leaving room for longer schedules or curriculum strategies. Truncating audio at 9 s throws away information in longer recordings. We rely on a single model with no ensembling. Finally, the OC-Softmax center vector may overspecialize to the dialects present in training, which would explain why so many out-of-distribution deepfakes receive confident bonafide scores.

6 Conclusion

We described a straightforward system - WavLM Large, LoRA, Attentive Statistics Pooling, and OC-Softmax - that beats the shared-task baseline on both tracks of ArA-DF 2026. The augmentation pipeline proves especially effective for acoustic robustness (Track 2 EER 5.66% vs. the baseline's 27.11%), but dialect generalization remains the harder problem. Going forward, the most promising directions seem to be dialect-aware training or adaptation strategies, better handling of very short clips (e.g., through overlapping chunks or duration-adaptive padding), and model ensembling.

References

Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, Jian Wu, Long Zhou, Shuo Ren, Yanmin Qian, Yao Qian, Jian Wu, Michael Zeng, Xiangyu Yu, and Furu Wei. 2022. Wavlm: Large-scale self-supervised pre-training for full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing*, 16(6):1505–1518.

Brecht Desplanques, Jenthe Thienpondt, and Kris Demuynck. 2020. ECAPA-TDNN: Emphasized channel attention, propagation and aggregation in TDNN based speaker verification. In *Proc. Interspeech*, pages 3830–3834.

Edward J Hu, yelong shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

Tom Ko, Vijayaditya Peddinti, Daniel Povey, Michael L. Seltzer, and Sanjeev Khudanpur. 2017. A study on data augmentation of reverberant speech for robust speech recognition. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 5220–5224. IEEE.

Janne Laakkonen, Ivan Kukanov, and Ville Hautamäki. 2025. Generalizable speech deepfake detection via meta-learned LoRA. *arXiv preprint arXiv:2502.10838*.

Koji Okabe, Takafumi Koshinaka, and Koichi Shinoda. 2018. Attentive statistics pooling for deep speaker embedding. In *Proc. Interspeech*, pages 2252–2256.

Daniel S. Park, William Chan, Yu Zhang, Chung-Cheng Chiu, Barret Zoph, Ekin D. Cubuk, and Quoc V. Le. 2019. SpecAugment: A Simple Data Augmentation Method for Automatic Speech Recognition. In *Interspeech 2019*, pages 2613–2617.

David Snyder, Guoguo Chen, and Daniel Povey. 2015. Musan: A music, speech, and noise corpus. *arXiv preprint arXiv:1510.08484*.

Chenglong Wang, Jiangyan Yi, Xiaohui Zhang, Jianhua Tao, Le Xu, and Ruibo Fu. 2023. Low-rank adaptation method for Wav2vec2-based fake audio detection. In *IJCAI 2023 Workshop on Deepfake Audio Detection and Analysis (DADA)*. ArXiv:2306.05617.

Xin Wang, Héctor Delgado, Hemlata Tak, Jee weon Jung, Hye jin Shim, Massimiliano Todisco, Ivan Kukanov, Xuechen Liu, Md Sahidullah, Tomi H. Kinnunen, Nicholas Evans, Kong Aik Lee, and Junichi Yamagishi. 2024. ASVspoof 5: crowdsourced speech data, deepfakes, and adversarial attacks at scale. In *The Automatic Speaker Verification Spoofing Countermeasures Workshop (ASVspoof 2024)*, pages 1–8.

Junichi Yamagishi, Xin Wang, Massimiliano Todisco, Md Sahidullah, Jose Patino, Andreas Nautsch, Xuechen Liu, Kong Aik Lee, Tomi Kinnunen, Nicholas Evans, and Héctor Delgado. 2021. ASVspoof 2021: accelerating progress in spoofed and deepfake speech detection. In *2021 Edition of the Automatic Speaker Verification and Spoofing Countermeasures Challenge*, pages 47–54.

You Zhang, Fei Jiang, and Zhiyao Duan. 2021. One-class learning towards synthetic voice spoofing detection. *IEEE Signal Processing Letters*, 28:937–941.